Divergence, Curl and Gradient Operator & Line, Surface, Volume Integral & Gauss Divergence and Stoke's Theorem

Maxwell's equations of electromagnetism

- All relationships between electric & magnetic fields & their sources summarized by four equations.

- James Clerk Maxwell

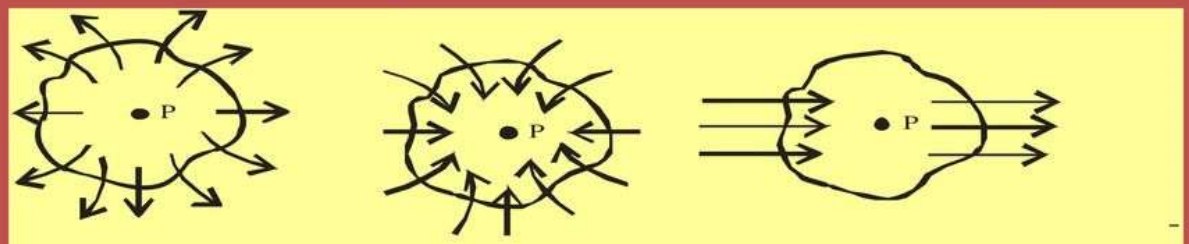
Divergence, Curl and Gradient Operations

(i) Divergence

- The divergence of a vector V written as div V represents the

div **V** = ∇ • **V** =

$$\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

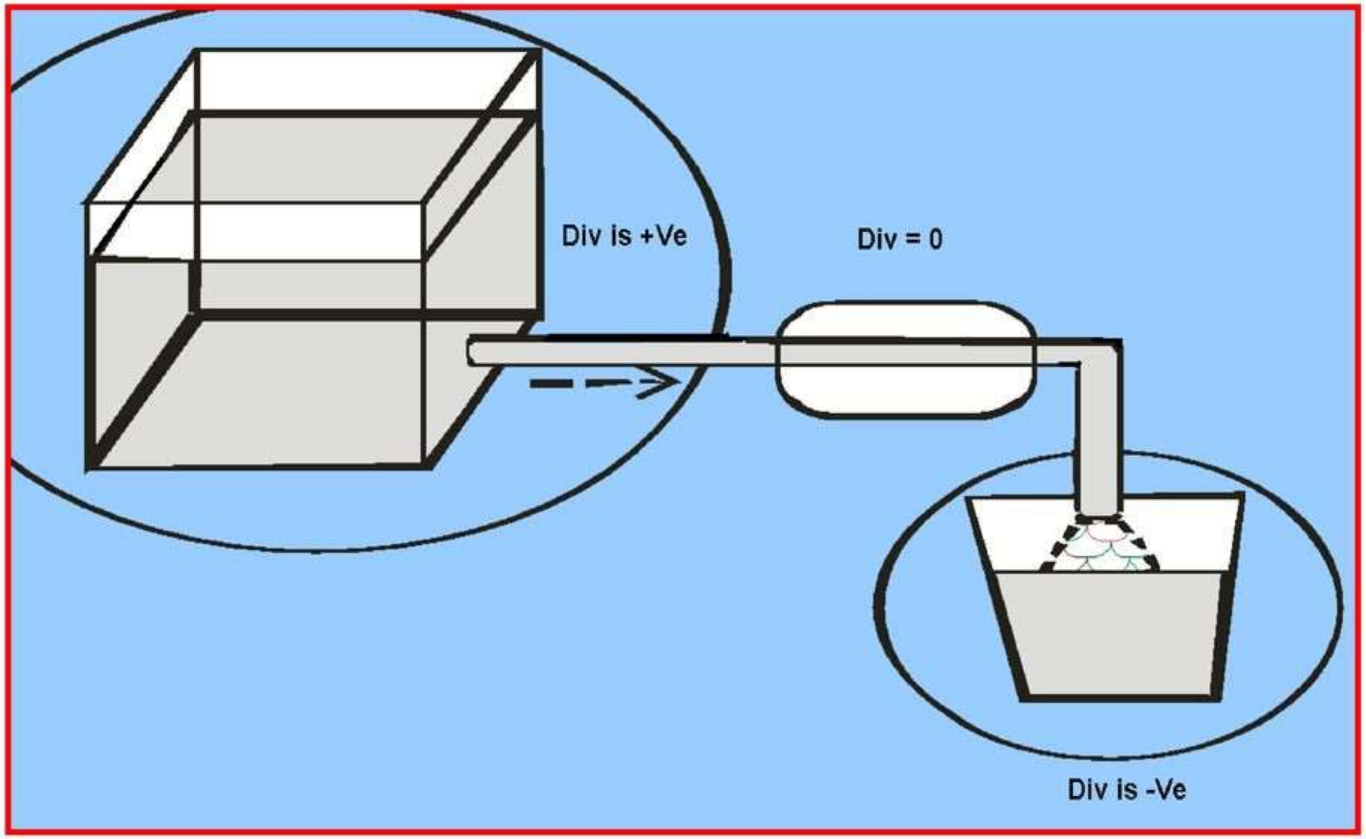


(a) positive divergence (b) negative divergence (c) zero divergence

Physical significance of divergence

- Physically the divergence of a vector quantity represents the rate of change of the field strength in the direction of the field.
- If the divergence of the vector field is positive at a point then something is diverging from a small volume surrounding with the point as a source.
- If it negative, then something is converging into the small volume surrounding that point is acting as sink.
- if the divergence at a point is zero then the rate at which something entering a small volume surrounding that point is equal to the rate at which it is leaving that volume.
- The vector field whose divergence is zero is called ***solenoidal***

Example for Divergence



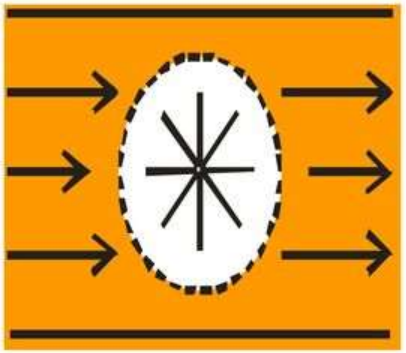
Curl of a Vector field

$$\text{Curl } \mathbf{V} = \nabla \times \vec{V} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix}$$

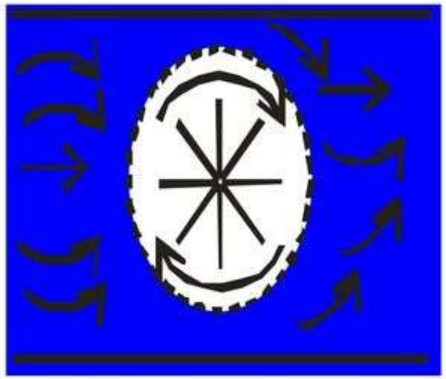
- Physically, the curl of a vector field represents the rate of change of the field strength in a direction at right angles to the field and is a measure of rotation of something in a small volume surrounding a particular point.

- For streamline motions and conservative fields, the curl is zero while it is maximum near the whirlpools

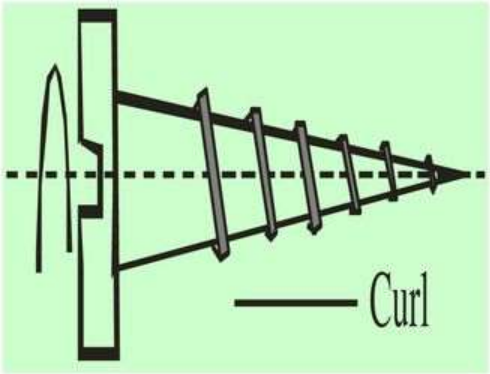
Representation of Curl



(I) No rotation of the paddle wheel represents zero curl



(II) Rotation of the paddle wheel showing the existence of curl



(III) direction of curl

➢For vector fields whose curl is zero there is no rotation of the paddle wheel when it is placed in the field, Such fields are called *irrotational*

A Cross, needless to say is made of two bars which are fixed perpendicular to each other.

Similarly, the 'X' symbol comes in maths whenever two forces produce maximum impact when kept perpendicular to each other.

But why does the Cross Come in the curl?

The value of Curl is maximum if you are perpendicular to the direction of the flow of field.

Gradient Operator

- The Gradient of a scalar function ϕ is a vector whose Cartesian components are,

$$\frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y} \text{ and } \frac{\partial \phi}{\partial z}$$

Then grad ϕ is given by,

$$\text{Grad}\phi = \nabla \phi = \vec{i}\frac{\partial \phi}{\partial x} + \vec{j}\frac{\partial \phi}{\partial y} + \vec{k}\frac{\partial \phi}{\partial z}$$

- The magnitude of this vector gives the maximum rate of change of the scalar field and directed towards the maximum change occurs.

- The electric field intensity at any point is given by,
  E* = – grad *V = negative gradient of potential

- The negative sign implies that the direction of *E* opposite to the direction in which *V* increases.

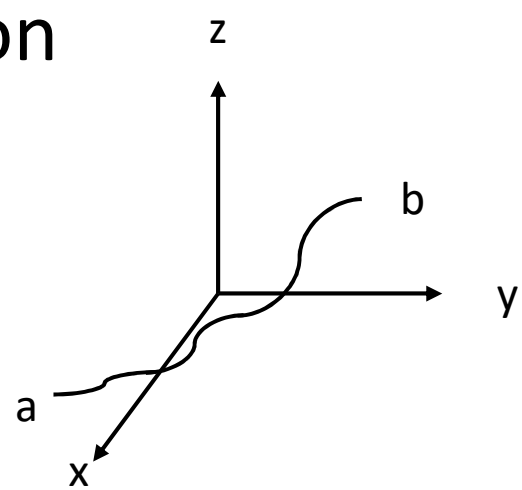
LINE INTEGRAL

- A line integral is an expression of the form

$$\int_a^b \overrightarrow{A} . \overrightarrow{dl}$$

$\overrightarrow{dl}$ - Infinitesimal displacement vector

A -Vector Function



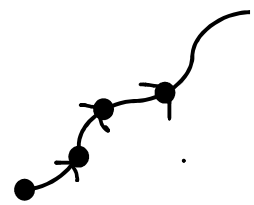
If you take line integral $\mathbf{a} \rightarrow \boldsymbol{b}$

If the path is closed one

$\oint \rightarrow$ Circle

$\oint \overrightarrow{\boldsymbol{A}}.\ \overrightarrow{\boldsymbol{dl}} \rightarrow$ Path is closed one

- At each point on the path we take a dot product of a, evaluated at that point with a displacement



- At each and every point on the path, we take a dot product of this vector with the displacement ‘dl’ to the next point on the path
- Eg : Work done by a force, $\vec{F}$
- **W = $\int \overrightarrow{F}.\ \overrightarrow{dl}$**

<u>Special case of vector function</u>

If line integral is independent of the path and it is determined only by the end points the force is conservative force.

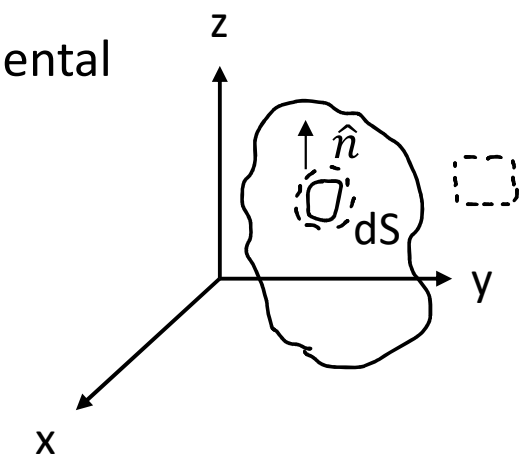
SURFACE INTEGRAL

- Surface integral of the form $\iint_s \vec{A}.\ \hat{n}\ \mathbf{ds}$
- $\hat{n}$ - is a unit vector

$\iint_s \vec{A}.\ \mathbf{ds}$

Consider a elemental surface and normal to the elemental surface is unit vector

Unit vector – length equal to 1 and perpendicular to the surface



ds = $\hat{n}$ ds

$\oiint_s \vec{A}.\ \vec{ds}$ - closed surface

'A' describes the flow of the field (mass per unit area per unit time)

$\iint_s \vec{A}.$ ds - Total mass per unit time or called flux

Eg : Current pass through a conductor

I = $\iint \vec{J}.\ \vec{ds}$ - Current flowing across unit area

VOLUME INTEGRAL

- A volume integral is an expression of the form

$\iiint_v \boldsymbol{\rho}\ \boldsymbol{dV}$ (ρ - Scalar function)

dV – Volume element

In cartesian coordinates dv = dx dy dz

Suppose ρ – represents charge density

$\iiint_v \rho\ dV$ - 'm'

This is volume integral $\frac{\boldsymbol{m}}{\boldsymbol{v}} = \boldsymbol{\rho}$

Gauss Divergence Theorem

- $\iiint_v (\vec{\nabla}.\vec{A})\, dV = \oiint_s \vec{A}.\, \vec{ds}$ Total outward flux
- Suppose if you consider a vector field 'A' . The volume integral of divergence of vector field 'A' is equal to the outward flux of the vector field , A, through the closed surface that bounds the volume.
- Gauss divergence theorem converts volume integral to surface integral and vice versa
- i.e. Volume integral of a vector field equals the closed surface integral of the vector field 'A'

Stoke's Theorem

- $\iint_s (\vec{\nabla} \times \vec{A})\, \vec{dS} = \oint_l \vec{A}.\, \vec{dl}$
- For a given vector field 'A', stoke's theorem states that the surface integral of the curl of the vector field, 'A' is an open surface is equal to the circulation of the vector field $\vec{A}$, along the boundary of the surface
- Stoke's theorem is the transformation from surface integral of curl of a vector to closed line integral of the vector and vice versa